

Education Experience:

I hold a BSc degree in Medical Physics with Physics from UCL. For my individual project, I focused on noninvasive electrophysiological signal, electromyography (EMG). This project provides me with experience of electronics, as I designed a bandpass filter circuit to remove artifacts before recording the EMG signals. Additionally, I utilized MATLAB for basic signal processing tasks. This project culminated in the creation of a prototype that translated the recorded EMG signal from my own arm into real-time sound tones. This creative project enhanced my problem-solving skills and deepened my interest in understanding and leveraging physiological data.

I am currently pursuing an MSc in Biomedical Engineering with a specialization in the Neurotechnology stream at Imperial College. This program has provided me with structured knowledge and research skills in the field of Neuroscience. I have gained a solid foundation in basic neuroscience principles and have learned about various methodologies for recording, processing, and analyzing neural data. In terms of programming, I have utilized both MATLAB and Python for tasks such as decoding spike data and modeling neural networks, specifically in the domain of Computational Neuroscience and Brain-Machine Interface.

For my ongoing MSc project, I am focusing on the study of brain organoids that are cultured from induced pluripotent stem cells. This in vitro research aims to track the neural dynamics across early-stage organoids with the goal of future applications in studying neurodegenerative diseases. The project involves hands-on work in the wet lab, where I take part in the growth and maintenance of cerebral organoids. Additionally, I will gain experience with two-photon microscopy during the data collection phase. The combination of my background knowledge, laboratory experience, and specialized training will be highly relevant and transferable to this applied work.

Additional Laboratory Experience:

During a summer internship, I had the opportunity to work in a biomedical ultrasound laboratory. My main task was to design experiments aimed at evaluating the performance of a launched ultrasound therapy system. This work differed from my previous experiences of being supervised on undergraduate and master's projects, as it demanded my decision-making skills and responsibility for the research progress. Over the course of two months, I gained valuable practical experience that would be beneficial for both my future academic studies and professional endeavors.

Motivation and Future Plan:

My initial interest in the field originated from my undergraduate project, which attempted to gather and decode information from EMG signals. This endeavor left me with lots of curiosity and many unanswered questions upon its completion. Motivated by the limitation of extracting only basic features at the time, I made the decision to delve into current advancements in neuroscience in order to pursue my passion.

This drove me to the current MSc program with Neurotechnology direction. The study of human brain and interpreting the human behavior and cognition is a complex puzzle. I believe that this opportunity could provide me with valuable knowledge and in-depth experience for applying a PhD position and starting my own research in the future.